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$$f_n : \mathbb{R} \to \mathbb{R} : x \to \begin{cases} \dfrac{\sqrt{n^2 - x^2}}{n} & \text{als} |x| \leq n \\ 0 & \text{elders} \end{cases}$$

$$\lim_{n\to\infty} \frac{\sqrt{n^2 - x^2}}{n} = \lim_{n\to\infty} \sqrt{1 - \frac{x^2}{n^2}} = 1$$

$$f : \mathbb{R} \to \mathbb{R} : x \to 1 \text{en} f_n \xrightarrow{p} f$$

$$||f_n - f||_\infty = \sup_{x\in\mathbb{R}} |f_n(x) - 1| = \sup_{|x|>n} |0 - 1| = 1 \neq 0$$

$$\Rightarrow f_n \overset{u}{\not\to} f$$

References